

BIP39 Printable Wordlist — Manual

These pages are meant to make rolling wallets or generating passphrases by hand easy to follow and as error-resistant as possible. Generate the bits, record them clearly, and use the lookup pages step by step.

Why

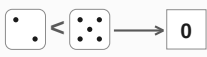
Software-based seed generation has a large attack surface: compromised RNGs, malware, clipboard sniffers, screen capture, or a backdoored hardware stack. Even an air-gapped computer still requires trusting its software and hardware chain.

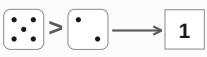
This layout lets you generate the seed words offline from your own physical randomness and look them up directly on paper. No screen or device is involved in the entropy-generation step.

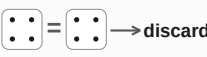
The layout is designed to keep rolling wallets or generating passphrases by hand simple, to reduce avoidable mistakes during recording and lookup, and to keep the random generation secure even when faced with low-quality dice.

Generating bits with a die

Use this bias-resistant method: always use a single die and roll that same die twice. If both values are equal, discard the pair. If the first value is smaller than the second, write 0. If the first value is greater than the second, write 1. If you discard a pair, start over with two fresh rolls.







Why the two-roll method is unbiased

$$p(a < b) = p(a)p(b) = p(a > b)$$

0

1

For any two different numbers a and b, getting a then b is just as likely as getting b then a. Shake well, and only use one die, so each roll is made the same way.

Expected number of rolls: for a fair d6, about 26.4 raw rolls per 11-bit word. For a fair d20, about 23.2. In general, only ties are discarded, so an n-sided die wastes just 1/n of pairs.

Using the lookup pages

For one BIP39 word, record 11 bits and split them as 2 | 3 | 3 = page | row | column | word.

page	1 2	bits 1–2	column	6 7 8	bits 6–8
row	3 4 5	bits 3–5	word	9 10 11	bits 9–11

Use the first 2 bits for the page, the next 3 for the row band, the next 3 for the column, and the last 3 for the word inside that cell.

The worksheet on the next page lets you record the page, row, column, and word bits, and also write down the actual BIP39 word.

How to create a wallet

Generate the words on paper first. Then open the restore/import flow on a trusted hardware wallet. This is the trick: do not let the wallet create a new phrase. Instead, enter the words you generated on paper as if the wallet already existed. In simple terms, restoring those words gives you the same wallet the device would have created if it had generated that exact phrase itself.

For most people, 12 words are enough when they are generated securely. Longer phrases are possible, but they are not required just to get strong security.

For a 12-word phrase, generate the first 11 words normally. For the 12th word, only the first 7 bits come from your own rolls. The last 4 bits are a BIP39 checksum. They are there for error detection, so wallets can reject mistyped or otherwise invalid phrases. On the worksheet, those are the greyed fields in row 12. So after you have fixed the first 7 bits, there are 16 possible final words that fit that pattern.

Try those final words one by one in the wallet's restore/import screen. In other words, you are using the restore screen to test which final phrase is valid. The wallet will reject the wrong ones and accept the right one. The accepted phrase is your real mnemonic.

Expect about 8 tries on average. Some wallets may make you enter all 12 words each time, so this can take quite some time.

The same idea works for longer phrases. On the worksheet, the greyed fields mark the bits you do not generate in the final row of a 12-, 15-, 18-, 21-, or 24-word phrase: 15 words → 32 candidates, 18 → 64, 21 → 128, 24 → 256. Avoid websites or extra software for checking candidate phrases unless they are fully offline and truly trusted.

Backing up the words

This project only helps with generation and lookup. Safe storage is a separate problem: paper vs. metal, geographic splits, memorization, inheritance planning, and so on. Lose the phrase and the funds are unrecoverable; leak it and someone else can spend them.

Additional detail: exact mapping

You do not need this to use the sheet. It only explains how the printed layout maps to the exact BIP39 index.

Read the 11 recorded bits from left to right. Split them as 2 | 3 | 3 | 3, which means page | row | column | word.

If you want the exact BIP39 list position, read the full 11-bit pattern as a binary number. The BIP39 wordlist counts from 0, so the Nth word uses binary(N-1).

Example: 1123th word

The 1123th BIP39 word is 'middle'. Take 1123-1 = 1122. In 11-bit binary that is 10001100010, or split up: 10 | 001 | 100 | 010.

1 0 | 0 0 1 | 1 0 0 | 0 1 0

BIP39 Printable Wordlist — Worksheet

Use this page to record the 11 bits for each word and the resulting BIP39 word itself. The worksheet is meant for the bias-resistant two-roll method.

Process

1. Use a single die and roll that same die twice. If both rolls are equal, discard that pair.
2. If the first roll is smaller than the second, write 0. If the first roll is greater than the second, write 1.
3. If you discard a pair, start over with two fresh rolls.
4. Fill the boxes from left to right: 2 bits page, 3 bits row, 3 bits column, 3 bits word.
5. Look up the resulting BIP39 word and write it in the last column.

Example filled row

1

0

001

100

010

→ middle

page

row

column

word

On rows 12, 15, 18, 21, and 24, the light grey boxes mark checksum bits in the final word. Do not roll those bits if that row is the last row for the number of words you want to use. Only generate the remaining boxes in that row, then find the accepted final word by recovery/import.

	page	row	column	word	actual word
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20					
21					
22					
23					
24					

	000	001	010	011	100	101	110	111
000	0000 abandon 0001 ability 0100 able 0111 about 1000 above 1001 absent 1100 absorb 1111 abstract	0000 absurd 0001 abuse 0100 access 0111 accident 1000 account 1001 accuse 1100 achieve 1111 acid	0000 acoustic 0001 acquire 0100 across 0111 act 1000 action 1001 actor 1100 actress 1111 actual	0000 adapt 0001 add 0100 addict 0111 address 1000 adjust 1001 admit 1100 adult 1111 advance	0000 advice 0001 aerobic 0100 affair 0111 afford 1000 afraid 1001 again 1100 age 1111 agent	0000 agree 0001 ahead 0100 aim 0111 air 1000 airport 1001 aisle 1100 alarm 1111 album	0000 alcohol 0001 alert 0100 alien 0111 all 1000 alley 1001 allow 1100 almost 1111 alone	0000 alpha 0001 already 0100 also 0111 alter 1000 always 1001 amateur 1100 amazing 1111 among
	0000 amount 0001 amused 0100 analyst 0111 anchor 1000 ancient 1001 anger 1100 angle 1111 angry	0000 animal 0001 ankle 0100 announce 0111 annual 1000 another 1001 answer 1100 antenna 1111 antique	0000 anxiety 0001 any 0100 apart 0111 apology 1000 appear 1001 apple 1100 approve 1111 april	0000 arch 0001 arctic 0100 area 0111 arena 1000 argue 1001 arm 1100 armed 1111 armor	0000 army 0001 around 0100 arrange 0111 arrest 1000 arrive 1001 arrow 1100 art 1111 artefact	0000 artist 0001 artwork 0100 ask 0111 aspect 1000 assault 1001 asset 1100 assist 1111 assume	0000 asthma 0001 athlete 0100 atom 0111 attack 1000 attend 1001 attitude 1100 attract 1111 auction	0000 audit 0001 august 0100 aunt 0111 author 1000 auto 1001 autumn 1100 average 1111 avocado
	0000 avoid 0001 awake 0100 aware 0111 away 1000 awesome 1001 awful 1100 awkward 1111 axis	0000 baby 0001 bachelor 0100 bacon 0111 badge 1000 bag 1001 balance 1100 balcony 1111 ball	0000 bamboo 0001 banana 0100 banner 0111 bar 1000 barely 1001 bargain 1100 barrel 1111 base	0000 basic 0001 basket 0100 battle 0111 beach 1000 bean 1001 beauty 1100 because 1111 become	0000 beef 0001 before 0100 begin 0111 behave 1000 behind 1001 believe 1100 below 1111 belt	0000 bench 0001 benefit 0100 best 0111 betray 1000 better 1001 between 1100 beyond 1111 bicycle	0000 bid 0001 bike 0100 bind 0111 biology 1000 bird 1001 birth 1100 bitter 1111 black	0000 blade 0001 blame 0100 blanket 0111 blast 1000 bleak 1001 bless 1100 blind 1111 blood
	0000 blossom 0001 blouse 0100 blue 0111 blur 1000 blush 1001 board 1100 boat 1111 body	0000 boil 0001 bomb 0100 bone 0111 bonus 1000 book 1001 boost 1100 border 1111 boring	0000 borrow 0001 boss 0100 bottom 0111 bounce 1000 box 1001 boy 1100 bracket 1111 brain	0000 brand 0001 brass 0100 brave 0111 bread 1000 breeze 1001 brick 1100 bridge 1111 brief	0000 bright 0001 bring 0100 brisk 0111 broccoli 1000 broken 1001 bronze 1100 broom 1111 brother	0000 brown 0001 brush 0100 bubble 0111 buddy 1000 budget 1001 buffalo 1100 build 1111 bulb	0000 bulk 0001 bullet 0100 bundle 0111 bunker 1000 burden 1001 burger 1100 burst 1111 bus	0000 business 0001 busy 0100 butter 0111 buyer 1000 buzz 1001 cabbage 1100 cabin 1111 cable
	0000 cactus 0001 cage 0100 cake 0111 call 1000 calm 1001 camera 1100 camp 1111 can	0000 canal 0001 cancel 0100 candy 0111 cannon 1000 canoe 1001 canvas 1100 canyon 1111 capable	0000 capital 0001 captain 0100 car 0111 carbon 1000 card 1001 cargo 1100 carpet 1111 carry	0000 cart 0001 case 0100 cash 0111 casino 1000 castle 1001 casual 1100 cat 1111 catalog	0000 catch 0001 category 0100 cattle 0111 caught 1000 cause 1001 caution 1100 cave 1111 ceiling	0000 celery 0001 cement 0100 census 0111 century 1000 cereal 1001 certain 1100 chair 1111 chalk	0000 champion 0001 change 0100 chaos 0111 chapter 1000 charge 1001 chase 1100 chat 1111 cheap	0000 check 0001 cheese 0100 chef 0111 cherry 1000 chest 1001 chicken 1100 chief 1111 child
	0000 chimney 0001 choice 0100 choose 0111 chronic 1000 chuckle 1001 chunk 1100 churn 1111 cigar	0000 cinnamon 0001 circle 0100 citizen 0111 city 1000 civil 1001 claim 1100 clap 1111 clarify	0000 claw 0001 clay 0100 clean 0111 clerk 1000 clever 1001 click 1100 client 1111 cliff	0000 climb 0001 clinic 0100 clip 0111 clock 1000 clog 1001 close 1100 cloth 1111 cloud	0000 clown 0001 club 0100 clump 0111 cluster 1000 clutch 1001 coach 1100 coast 1111 coconut	0000 code 0001 coffee 0100 coil 0111 coin 1000 collect 1001 color 1100 column 1111 combine	0000 come 0001 comfort 0100 comic 0111 common 1000 company 1001 concert 1100 conduct 1111 confirm	0000 congress 0001 connect 0100 consider 0111 control 1000 convince 1001 cook 1100 cool 1111 copper
	0000 copy 0001 coral 0100 core 0111 corn 1000 correct 1001 cost 1100 cotton 1111 couch	0000 country 0001 couple 0100 course 0111 cousin 1000 cover 1001 coyote 1100 crack 1111 cradle	0000 craft 0001 cram 0100 crane 0111 crash 1000 crater 1001 crawl 1100 crazy 1111 cream	0000 credit 0001 creek 0100 crew 0111 cricket 1000 crime 1001 crisp 1100 critic 1111 crop	0000 cross 0001 crouch 0100 crowd 0111 crucial 1000 cruel 1001 cruise 1100 crumble 1111 crunch	0000 crush 0001 cry 0100 crystal 0111 cube 1000 culture 1001 cup 1100 cupboard 1111 curious	0000 current 0001 curtain 0100 curve 0111 cushion 1000 custom 1001 cute 1100 cycle 1111 dad	0000 damage 0001 damp 0100 dance 0111 danger 1000 daring 1001 dash 1100 daughter 1111 dawn
	0000 day 0001 deal 0100 debate 0111 debris 1000 decade 1001 december 1100 decide 1111 decline	0000 decorate 0001 decrease 0100 deer 0111 defense 1000 define 1001 defy 1100 degree 1111 delay	0000 deliver 0001 demand 0100 demise 0111 denial 1000 dentist 1001 deny 1100 depart 1111 depend	0000 deposit 0001 depth 0100 deputy 0111 derive 1000 describe 1001 desert 1100 design 1111 desk	0000 despair 0001 destroy 0100 detail 0111 detect 1000 develop 1001 device 1100 devote 1111 diagram	0000 dial 0001 diamond 0100 diary 0111 dice 1000 diesel 1001 diet 1100 differ 1111 digital	0000 dignity 0001 dilemma 0100 dinner 0111 dinosaur 1000 direct 1001 dirt 1100 disagree 1111 discover	0000 disease 0001 dish 0100 dismiss 0111 disorder 1000 display 1001 distance 1100 divert 1111 divide

	000	001	010	011	100	101	110	111
000	000divorce 001dizzy 010doctor 011document 100dog 101doll 110dolphin 111domain	000donate 001donkey 010donor 011door 100dose 101double 110dove 111draft	000dragon 001drama 010drastic 011draw 100dream 101dress 110drift 111drill	000drink 001drip 010drive 011drop 100drum 101dry 110duck 111dumb	000dune 001during 010dust 011dutch 100duty 101dwarf 110dynamic 111eager	000eagle 001early 010earn 011earth 100easily 101east 110easy 111echo	000ecology 001economy 010edge 011edit 100educate 101effort 110egg 111eight	000either 001elbow 010elder 011electric 100elegant 101element 110elephant 111elevator
001	000elite 001else 010embark 011embody 100embrace 101emerge 110emotion 111employ	000empower 001empty 010enable 011enact 100end 101endless 110endorse 111enemy	000energy 001enforce 010engage 011engine 100enhance 101enjoy 110enlist 111enough	000enrich 001enroll 010ensure 011enter 100entire 101entry 110envelope 111episode	000equal 001equip 010era 011erase 100erode 101erosion 110error 111erupt	000escape 001essay 010essence 011estate 100eternal 101ethics 110evidence 111evil	000evoke 001evolve 010exact 011example 100excess 101exchange 110excite 111exclude	000excuse 001execute 010exercise 011exhaust 100exhibit 101exile 110exist 111exit
010	000exotic 001expand 010expect 011expire 100explain 101expose 110express 111extend	000extra 001eye 010eyebrow 011fabric 100face 101faculty 110fade 111faint	000faith 001fall 010false 011fame 100family 101famous 110fan 111fancy	000fantasy 001farm 010fashion 011fat 100fatal 101father 110fatigue 111fault	000favorite 001feature 010february 011federal 100fee 101feed 110feel 111female	000fence 001festival 010fetch 011fever 100few 101fiber 110fiction 111field	000figure 001file 010film 011filter 100final 101find 110fine 111finger	000finish 001fire 010firm 011first 100fiscal 101fish 110fit 111fitness
011	000fix 001flag 010flame 011flash 100flat 101flavor 110flee 111flight	000flip 001float 010flock 011floor 100flower 101fluid 110flush 111fly	000foam 001focus 010fog 011foil 100fold 101follow 110food 111foot	000force 001forest 010forget 011fork 100fortune 101forum 110forward 111fossil	000foster 001found 010fox 011fragile 100frame 101frequent 110fresh 111friend	000fringe 001frog 010front 011frost 100frown 101frozen 110fruit 111fuel	000fun 001funny 010furnace 011fury 100future 101gadget 110gain 111galaxy	000gallery 001game 010gap 011garage 100garbage 101garden 110garlic 111garment
100	000gas 001gasp 010gate 011gather 100gauge 101gaze 110general 111genius	000genre 001gentle 010genuine 011gesture 100ghost 101giant 110gift 111giggle	000ginger 001giraffe 010girl 011give 100glad 101glance 110glare 111glass	000glide 001glimpse 010globe 011gloom 100glory 101glove 110glow 111glue	000goat 001goddess 010gold 011good 100goose 101gorilla 110gospel 111gossip	000govern 001gown 010grab 011grace 100grain 101grant 110grape 111grass	000gravity 001great 010green 011grid 100grief 101grit 110grocery 111group	000grow 001grunt 010guard 011guess 100guide 101guilt 110guitar 111gun
101	000gym 001habit 010hair 011half 100hammer 101hamster 110hand 111happy	000harbor 001hard 010harsh 011harvest 100hat 101have 110hawk 111hazard	000head 001health 010heart 011heavy 100hedgehog 101height 110hello 111helmet	000help 001hen 010hero 011hidden 100high 101hill 110hint 111hip	000hire 001history 010hobby 011hockey 100hold 101hole 110holiday 111hollow	000home 001honey 010hood 011hope 100horn 101horror 110horse 111hospital	000host 001hotel 010hour 011hover 100hub 101huge 110human 111humble	000humor 001hundred 010hungry 011hunt 100hurdle 101hurry 110hurt 111husband
110	000hybrid 001ice 010icon 011idea 100identify 101idle 110ignore 111ill	000illegal 001illness 010image 011imitate 100immense 101immune 110impact 111impose	000improve 001impulse 010inch 011include 100income 101increase 110index 111indicate	000indoor 001industry 010infant 011inflict 100inform 101inhale 110inherit 111initial	000inject 001injury 010inmate 011inner 100innocent 101input 110inquiry 111insane	000insect 001inside 010inspire 011install 100intact 101interest 110into 111invest	000invite 001involve 010iron 011island 100isolate 101issue 110item 111ivory	000jacket 001jaguar 010jar 011jazz 100jealous 101jeans 110jelly 111jewel
111	000job 001join 010joke 011journey 100joy 101judge 110juice 111jump	000jungle 001junior 010junk 011just 100kangaroo 101keen 110keep 111ketchup	000key 001kick 010kid 011kidney 100kind 101kingdom 110kiss 111kit	000kitchen 001kite 010kitten 011kiwi 100knee 101knife 110knock 111know	000lab 001label 010labor 011ladder 100lady 101lake 110lamp 111language	000laptop 001large 010later 011latin 100laugh 101laundry 110lava 111law	000lawn 001lawsuit 010layer 011lazy 100leader 101leaf 110learn 111leave	000lecture 001left 010leg 011legal 100legend 101leisure 110lemon 111lend

	000	001	010	011	100	101	110	111
000	length lens leopard lesson letter level liar liberty	library license life lift light like limb limit	link lion liquid list little live lizard load	loan lobster local lock logic lonely long loop	lottery loud lounge love loyal lucky luggage lumber	lunar lunch luxury lyrics machine mad magic magnet	maid mail main major make mammal man manage	mandate mango mansion manual maple marble march margin
001	marine market marriage mask mass master match material	math matrix matter maximum maze meadow mean measure	meat mechanic medal media melody melt member memory	mention menu mercy merge merit merry mesh message	metal method middle midnight milk million mimic mind	minimum minor minute miracle mirror misery miss mistake	mix mixed mixture mobile model modify mom moment	monitor monkey monster month moon moral more morning
010	mosquito mother motion motor mountain mouse move movie	much muffin mule multiply muscle museum mushroom music	must mutual myself mystery myth naive name napkin	narrow nasty nation nature near neck need negative	neglect neither nephew nerve nest net network neutral	never news next nice night noble noise nominee	noodle normal north nose notable note nothing notice	novel now nuclear number nurse nut oak obey
011	object oblige obscure observe obtain obvious occur ocean	october odor off offer office often oil okay	old olive olympic omit once one onion online	only open opera opinion oppose option orange orbit	orchard order ordinary organ orient original orphan ostrich	other outdoor outer output outside oval oven over	own owner oxygen oyster ozone pact paddle page	pair palace palm panda panel panic panther paper
100	parade parent park parrot party pass patch path	patient patrol pattern pause pave payment peace peanut	pear peasant pelican pen penalty pencil people pepper	perfect permit person pet phone photo phrase physical	piano picnic picture piece pig pigeon pill pilot	pink pioneer pipe pistol pitch pizza place planet	plastic plate play please pledge pluck plug plunge	poem poet point polar pole police pond pony
101	pool popular portion position possible post potato pottery	poverty powder power practice praise predict prefer prepare	present pretty prevent price pride primary print priority	prison private prize problem process produce profit program	project promote proof property prosper protect proud provide	public pudding pull pulp pulse pumpkin punch pupil	puppy purchase purity purpose purse push put puzzle	pyramid quality quantum quarter question quick quit quiz
110	quote rabbit raccoon race rack radar radio rail	rain raise rally ramp ranch random range rapid	rare rate rather raven raw razor ready real	reason rebel rebuild recall receive recipe record recycle	reduce reflect reform refuse region regret regular reject	relax release relief rely remain remember remind remove	render renew rent reopen repair repeat replace report	require rescue resemble resist resource response result retire
111	retreat return reunion reveal review reward rhythm rib	ribbon rice rich ride ridge rifle right rigid	ring riot ripple risk ritual rival river road	roast robot robust rocket romance roof rookie room	rose rotate rough round route royal rubber rude	rug rule run runway rural sad saddle sadness	safe sail salad salmon salon salt salute same	sample sand satisfy satoshi sauce sausage save say

	000	001	010	011	100	101	110	111
000	000scale 001scan 010scare 011scatter 100scene 101scheme 110school 111science	000scissors 001scorpion 010scout 011scrap 100screen 101script 110scrub 111sea	000search 001season 010seat 011second 100secret 101section 110security 111seed	000seek 001segment 010select 011sell 100seminar 101senior 110sense 111sentence	000series 001service 010session 011settle 100setup 101seven 110shadow 111shaft	000shallow 001share 010shed 011shell 100sheriff 101shield 110shift 111shine	000ship 001shiver 010shock 011shoe 100shoot 101shop 110short 111shoulder	000shove 001shrimp 010shrug 011shuffle 100shy 101sibling 110sick 111side
001	000siege 001sight 010sign 011silent 100silk 101silly 110silver 111similar	000simple 001since 010sing 011siren 100sister 101situate 110six 111size	000skate 001sketch 010ski 011skill 100skin 101skirt 110skull 111slab	000slam 001sleep 010slender 011slice 100slide 101slight 110slim 111slogan	000slot 001slow 010slush 011small 100smart 101smile 110smoke 111smooth	000snack 001snake 010snap 011sniff 100snow 101soap 110soccer 111social	000sock 001soda 010soft 011solar 100soldier 101solid 110solution 111solve	000someone 001song 010soon 011sorry 100sort 101soul 110sound 111soup
010	000source 001south 010space 011spare 100spatial 101spawn 110speak 111special	000speed 001spell 010spend 011sphere 100spice 101spider 110spike 111spin	000spirit 001split 010spoil 011sponsor 100spoon 101sport 110spot 111spray	000spread 001spring 010spy 011square 100squeeze 101squirrel 110stable 111stadium	000staff 001stage 010stairs 011stamp 100stand 101start 110state 111stay	000steak 001steel 010stem 011step 100stereo 101stick 110still 111sting	000stock 001stomach 010stone 011stool 100story 101stove 110strategy 111street	000strike 001strong 010struggle 011student 100stuff 101stumble 110style 111subject
011	000submit 001subway 010success 011such 100sudden 101suffer 110sugar 111suggest	000suit 001summer 010sun 011sunny 100sunset 101super 110supply 111supreme	000sure 001surface 010surge 011surprise 100surround 101survey 110suspect 111sustain	000swallow 001swamp 010swap 011swarm 100swear 101sweet 110swift 111swim	000swing 001switch 010sword 011symbol 100symptom 101syrup 110system 111table	000tackle 001tag 010tail 011talent 100talk 101tank 110tape 111target	000task 001taste 010tattoo 011taxi 100teach 101team 110tell 111ten	000tenant 001tennis 010tent 011term 100test 101text 110thank 111that
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101	000trend 001trial 010tribe 011trick 100trigger 101trim 110trip 111trophy	000trouble 001truck 010true 011truly 100trumpet 101trust 110truth 111try	000tube 001tuition 010tumble 011tuna 100tunnel 101turkey 110turn 111turtle	000twelve 001twenty 010twice 011twin 100twist 101two 110type 111typical	000ugly 001umbrella 010unable 011unaware 100uncle 101uncover 110under 111undo	000unfair 001unfold 010unhappy 011uniform 100unique 101unit 110universe 111unknown	000unlock 001until 010unusual 011unveil 100update 101upgrade 110uphold 111upon	000upper 001upset 010urban 011urge 100usage 101use 110used 111useful
110	000useless 001usual 010utility 011vacant 100vacuum 101vague 110valid 111valley	000valve 001van 010vanish 011vapor 100various 101vast 110vault 111vehicle	000velvet 001vendor 010venture 011venue 100verb 101verify 110version 111very	000vessel 001veteran 010viable 011vibrant 100vicious 101victory 110video 111view	000village 001vintage 010violin 011virtual 100virus 101visa 110visit 111visual	000vital 001vivid 010vocal 011voice 100void 101volcano 110volume 111vote	000voyage 001wage 010wagon 011wait 100walk 101wall 110walnut 111want	000warfare 001warm 010warrior 011wash 100wasp 101waste 110water 111wave
111	000way 001wealth 010weapon 011wear 100weasel 101weather 110web 111wedding	000weekend 001weird 010welcome 011west 100wet 101whale 110what 111wheat	000wheel 001when 010where 011whip 100whisper 101wide 110width 111wife	000wild 001will 010win 011window 100wine 101wing 110wink 111winner	000winter 001wire 010wisdom 011wise 100wish 101witness 110wolf 111woman	000wonder 001wood 010wool 011word 100work 101world 110worry 111worth	000wrap 001wreck 010wrestle 011wrist 100write 101wrong 110yard 111year	000yellow 001you 010young 011youth 100zebra 101zero 110zone 111zoo